About 2,600 results for “review wildfire spread prediction”

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Methodological foundations of forecasting the spread of diseases in the world (review)

V. O. Shchepin | A. Zagoruychenko | O. B. Karpova | Medicine · Manager Zdravoochranenia · 1 September 2022

TLDR The analysis of the experience of predicting morbidity made it possible to determine the features of the use of certain mixed methods of forecasting different classes, groups and types of diseases, especially when identifying new viruses and infectious diseases. Expand

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Comparing Satellite Data for Next-Day Wildfire Predictability

Justus Karlsson | Yonghao Xu | Amanda Berg | Leif Haglund | Environmental Science · arXiv.org · 2025

—Multiple studies have performed next-day fire **prediction** using satellite imagery. Two main satellites are used to detect **wildfires**: MODIS and VIIRS. Both satellites provide fire mask products,… Expand

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Comparing Next-Day Wildfire Predictability of MODIS and VIIRS Satellite Data

Justus Karlsson | Yonghao Xu | Amanda Berg | Leif Haglund | Environmental Science · Scandinavian Conference on Image Analysis · 11 March 2025

Multiple studies have performed next-day fire **prediction** using satellite imagery. Two main satellites are used to detect **wildfires**: MODIS and VIIRS. Both satellites provide fire mask products, called… Expand

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Spatial Uncertainty Quantification in Wildfire Forecasting for Climate-Resilient Emergency Planning

Aditya Chakravarty | Environmental Science, Computer Science · 8 October 2025

TLDR It is demonstrated that predictive uncertainty exhibits coherent spatial structure concentrated near fire perimeters, and a novel distance metric reveals high-uncertainty regions form consistent 20-60 meter buffer zones around predicted firelines - directly applicable for emergency planning. Expand

[PDF] | arXiv | Save | Cite

Pre-Ignition Wildfire Burn Severity Modeling with Polarimetric UAVSAR Data and Machine Learning

Kris Mannino | D. Deb | Karen An | Environmental Science, Engineering · SoutheastCon · 22 March 2025

TLDR This research presents a model of burn severity **prediction** based on pre-fire static variables with correlations to burn severity using data available pre-ignition for insight of **wildfire** propagation within a known ignition perimeter. Expand

IEEE | Save | Cite

Using Wind Data to Predict Wildfire Spread in Central California

F. Du | R. Rothermel | Environmental Science · 2017

Wildfire growth simulation is important for identifying and mitigating potential **wildfires**, distributing firefighting resources, and understanding past fire incidents. During **wildfire** growth…

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A MACROECONOMIC MODEL FOR CREDIT SPREAD PREDICTION IN CHINA'S CORPORATE BOND MARKET

Y. Yue Zheng Guipeng Yining Long Z. Liang Economics • 2014

of variations of the credit **spreads** observed from the 1 year YTM of AAA and AA corporate bonds in China are 0.22 and 0.64, respectively, meaning that credit **spread** variations account for 64% of the... Expand

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Radiomics for Identification and Prediction in Metastatic Prostate Cancer: A Review of Studies

J. Kendrick R. Francis +5 authors M. Ebert Medicine, Engineering • Frontiers in Oncology • 1 November 2021

TLDR An overview of the radiomics methodology is provided, detailing critical aspects affecting the reproducibility of features, and providing examples of how artificial intelligence techniques can be incorporated into the workflow. Expand

31 PDF PubMed Save Cite

Geo-referenced image data assimilation for wildfire spread simulation

Feng Gu R. Syeda Chunyu Ai Computer Science, Environmental Science • Spring Simulation Multiconference • 3 April 2016

TLDR This paper proposes the data assimilation method to assimilate the geo-referenced images into the **wildfire spread** simulation using sequential Monte Carlo (SMC) methods and shows its effectiveness to improve the state estimation through the designed experiments. Expand

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Evaluating the performance of WRF in simulating winds and surface meteorology during a Southern California wildfire event

Mukesh Kumar B. Kosović Hara P. Nayak William C. Porter J. Randerson T. Banerjee

Environmental Science • Frontiers in Earth Science • 19 January 2024

The intensity and frequency of **wildfires** in California (CA) have increased in recent years, causing significant damage to human health and property. In October 2007, a number of small fire events,... Expand

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